

tion multimedia interfaces (HDMI) so as to find place in an ideal home theatre environment.

"Typically, home theatre projectors come with native resolutions of 720p (720 horizontal, progressive scan lines) and 1080p, an aspect ratio of 16:9 and higher contrast ratios. Acer's Full HD 1080p offers full native 1080p wide-screen resolution that delivers a true cinematic experience for the latest high-definition content. 1080p is a perfect fit for wide-screen movies created in Blu-ray disk and high-definition television (HDTV) formats, as the content mirrors the original creation and every single detail is perfectly preserved with true-to-master precision," says S. Rajendran, chief marketing officer, Acer India.

### Into 3D

There is more to projection than just high-definition video. Even as the television world shows signs of introducing 3D viewing with a few brands offering 3D-ready HDTVs and some global service providers planning 3D channels, the projector market is already marching in pace.

"3D-ready models are ideal for business and education applications where 3D rendering is important to deliver the message well. It is great for home theatre installation and compatible with over 350 3D video games," comments Ghosh of ViewSonic, which offers 3D-ready models such as the PJ6 series of projectors with 120Hz frame rate to ensure blur-free images.

Acer too has recently introduced a 3D-ready DLP projector. The X1261 model uses millions of microscopic, digital mirrors that reflect light to create a stunning picture. "This imaging technology is so fast that it can actually produce two images on the screen at the same time: one for the left eye and one for the right eye using a single projector. Then 3D glasses combine the two images to create an amazing 3D effect," explains Rajendran.

At InfoComm 2009, Texas Instruments (TI) announced that it would

launch affordable 3D DLP projectors made specifically for the classroom environment. According to the company, 3D projectors will soon be available at the price of non-3D eXtended Graphics Array (XGA) and Super XGA projectors. And non-3D projectors will also be able to upgrade to 3D via firmware updates.

Like Acer projectors, TI's 3D projectors too will use only one projector coupled with 'DLP Link' active shutter glasses. No infrared emitters, polarisers or special screens are required. The technology works by dividing the projector's 120Hz output between the left and right eyes (60 Hz each), along with synchronisation data coming through during ultra-brief 'dark' times between active data transmission. The required glasses will also be available at a low cost. Plus, the 3D projectors are capable of 2D projection.

Other companies offering 3D-ready projectors include Optoma, Mitsubishi and BenQ.

### Game to network, even wirelessly

Some high-end projector models now support Internet connectivity, local-area network connectivity and/or wireless connectivity.

Network-capable projectors usually come with Ethernet connections that enable them to connect to the local-area or wide-area network. This allows not only data-sharing but also remote diagnostics and management of the device, just like other networked resources such as printers. Epson, Hitachi, BenQ, Sony, Dell and most other companies offer network-ready projectors.

High-end models such as those from Epson and NEC can connect wirelessly. You can even connect the projector to multiple laptops at the same time.

A step ahead, Rajendran remarks,



Acer X1261

"Technology is always developing in different ways to aid the user. Network projection technology allows distance presentation from anywhere in the world with an Internet protocol (IP) network connection."

### With data-capable projectors, who needs a PC?

If all you need is to make a presentation, do you have to carry your laptop? May be not. Projectors are becoming more independent now, capable of directly playing presentations from USB drives, CDs, DVDs and other supported media.

Models from various companies including Epson, NEC and Toshiba offer support for various media, plus PC-less operation in several cases.

Any device that is capable of handling data as well as networking requires security features as well – to avoid unauthorised access, leakage of data, *et al.* Projectors are no different. Most companies offering data-capable and network-enabled projectors also include related features such as PIN or other forms of authorisation, easy setup and identification for networking purposes, and so on.

### Power bills? No more shocks

Energy-efficiency, and thereby power saving, is one of the main features of today's projectors. In fact, it is a key parameter considered by corporate buyers. Cooler operation (thereby